

Statistics Exam Master Guide

Multiple Linear Regression & Time Series

How to read questions · Which method to use · Why it works Common pitfalls ·
Different phrasings · Worked decision logic

This guide walks you through every type of question in exams on MLR and Time Series. Each section gives: what the question is really asking, which method to choose and why, step-by-step decision logic, the correct interpretation, actual plots to help you recognize patterns, pitfalls that cost marks, and different phrasings that mean the same thing.

PART 1 — MULTIPLE LINEAR REGRESSION (MLR)

1.1 How to Read an MLR Question

Read the question twice and identify these four things before opening SPSS:

Identify	Appears as	Action
Dependent variable Y	"Regress X on Y" / "explain Y" / "Y as a function of"	Dependent box in SPSS
Continuous predictor	Age, income, price — no category labels	Enter directly as IV
Categorical predictor	"Region", "type" with labels like North/South	Create k–1 dummy variables
Log transformation	"ln(X)", "log(X)" in the variable name	Already logged, or Transform → Compute
Interaction term	"effect depends on Z" / "X×Z"	Create product variable X×Z

1.2 Dummy Variables

A dummy variable converts a categorical variable with k categories into k–1 binary (0/1) variables. The excluded category is the reference group — every coefficient is interpreted relative to it.

Categories	Dummies to create	Reference (omitted)
North, South, Center, East (4)	D_North, D_South, D_East (3 dummies)	Center
Male, Female (2)	D_Female (1 dummy)	Male
Low, Medium, High (3)	D_Medium, D_High (2 dummies)	Low

SPSS: Transform → Compute Variable for each category (except reference):

`D_North = (Region = 1) [if North is coded 1 in data]`

`D_South = (Region = 2) [skip 3 = Center → it is the reference]`

`D_East = (Region = 4)`

Theoretical model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 \cdot D_{\text{North}} + \beta_3 \cdot D_{\text{South}} + \beta_4 \cdot D_{\text{East}} + \varepsilon$$

Interpretation of β_2 : North regions have β_2 units more Y than Center,

holding all other variables constant.

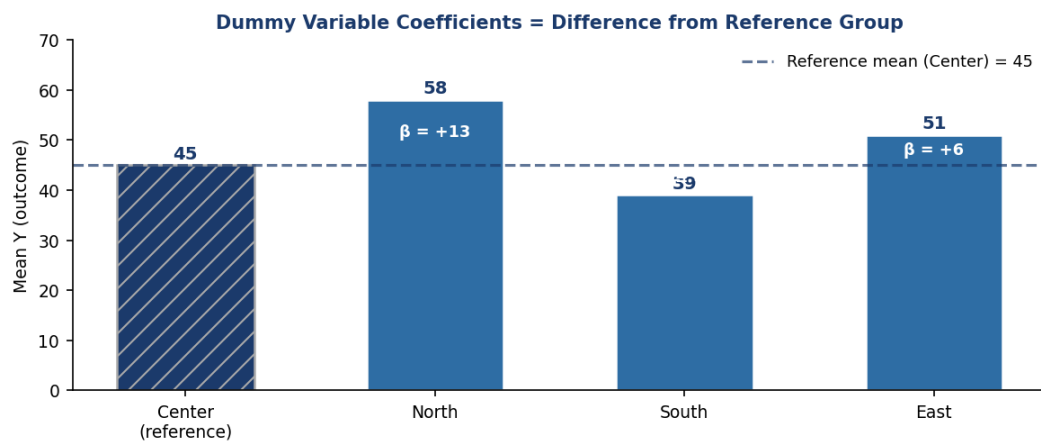


Figure 1.2 — Each bar shows the mean Y for that region. The dummy coefficient = the bar's height minus the reference bar (Center). The reference group has no dummy and its effect is absorbed into the intercept β_0 .

SPSS: Creating and Using Dummy Variables

1. Transform → Compute Variable...
2. Target Variable: D_North | Numeric Expression: (Region = 1) [use the numeric code for North]
3. Click OK. Repeat for each category EXCEPT the reference (e.g. Center).
4. Analyze → Regression → Linear...
5. Dependent: Y | Independent(s): X1, D_North, D_South, D_East
6. Click OK. Read Coefficients table: B column = estimates, Sig. column = p-values.

Common Pitfalls

- Never create k dummies — always k-1. Including all k causes perfect multicollinearity (dummy trap).
- "Baseline group", "comparison group", "omitted category", "reference category" — all the same.
- "Use Center as reference" means do NOT create a dummy for Center.

1.3 Log Transformations and Interpretation

Model	Equation	β interpretation
Linear	$Y = \alpha + \beta X + \varepsilon$	1-unit rise in X → β unit change in Y
Linear-log	$Y = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in X → $\beta/100$ unit change in Y
Log-linear	$\ln(Y) = \alpha + \beta X + \varepsilon$	1-unit rise in X → $\beta \times 100$ % change in Y
Log-log	$\ln(Y) = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in X → $\beta\%$ change in Y (elasticity)

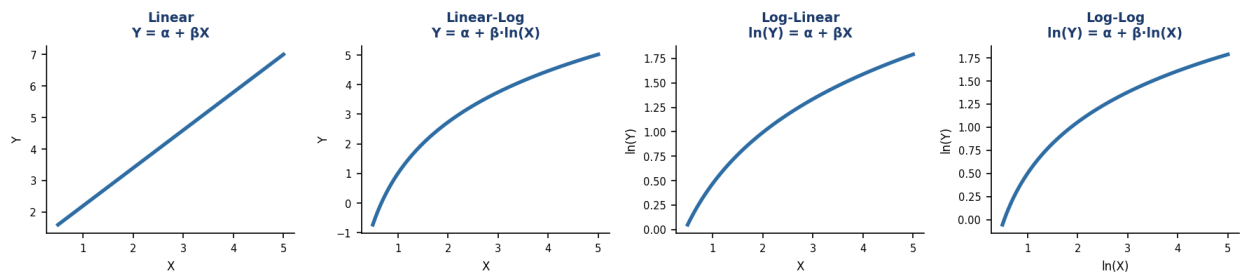


Figure 1.3 — Shape of each model type. Linear: straight line. Linear-log: curved (diminishing returns). Log-linear: exponential. Log-log: power curve (constant elasticity). Recognise which shape fits the data to identify the correct model type.

Most Tested: Linear-Log

- Most common exam type: Linear-Log. $Y = \alpha + \beta \ln(X) + \epsilon$
- Example: $\beta = -16.04$, X is $\ln(\text{NOx})$.
- Answer: 1% increase in $\text{NOx} \rightarrow -16.04/100 = -0.16$ units of Y . If Y is in thousands of dollars: $-\$160$ per 1% rise in pollution.

SPSS: Log Transformations

1. If the variable is NOT yet logged: Transform → Compute Variable...
2. Target Variable: LN_X | Numeric Expression: $\text{LN}(X)$ | Click OK.
3. If both sides need logging (log-log model): also create $\text{LN_Y} = \text{LN}(Y)$.
4. Analyze → Regression → Linear...
5. Dependent: Y (or LN_Y) | Independent(s): LN_X (and other predictors)
6. Click OK. Read B column for β – apply the correct interpretation rule from the table above.

1.4 Interpreting Coefficients — Decision Table

X logged?	Y logged?	Interpretation of β
No	No	1 extra unit of $X \rightarrow \beta$ extra units of Y (ceteris paribus)
Yes ($\ln X$)	No	1% extra in $X \rightarrow \beta/100$ extra units of Y
No	Yes ($\ln Y$)	1 extra unit of $X \rightarrow \beta \times 100$ % change in Y
Yes ($\ln X$)	Yes ($\ln Y$)	1% extra in $X \rightarrow \beta\%$ change in Y
Dummy (0/1)	No	Being in group D $\rightarrow \beta$ extra units of Y vs. reference

1.5 Testing Individual Coefficient Significance

$H_0: \beta = 0$ (variable has no effect on Y)

$H_1: \beta \neq 0$ (variable does affect Y)

Decision: p-value (Sig. column in SPSS) < 0.05 → Reject H_0 → significant at 5%

p-value ≥ 0.05 → Fail to reject H_0 → not significant

SPSS: Testing Individual Coefficient Significance

1. Analyze → Regression → Linear...
2. Dependent: Y | Independent(s): all predictor variables
3. Click OK. Open the Coefficients table in the output.
4. Find the row for the variable of interest.
5. Read the Sig. column (= p-value). If Sig. < 0.05 → significant at 5%.
6. Also read B (estimate) and t columns to quote in your answer.

Common Pitfalls

- "Significant" means statistically distinguishable from zero — NOT necessarily large or important.
- "Is X relevant?", "Does X matter?", "Test the effect of X", "Is β significantly different from zero?" — all ask for the t-test p-value.

1.6 Joint F-test — Testing a Group of Variables Together

Use joint F-test when you see these phrases

- "Are the region dummies jointly significant?"
- "Do the seasonal dummies together improve the model?"
- "Test whether D_1, D_2, D_3 together add explanatory power."
- Any question testing MORE THAN ONE variable simultaneously.

Step	What you do	Why
1	Run UNRESTRICTED model — all variables included	Gives SSE_u and R^2_u
2	Run RESTRICTED model — remove the variables being tested	Gives SSE_r and R^2_r
3	J = number of variables removed	J = numerator df
4	Read n (obs) and k (predictors in unrestricted model)	For denominator df
5	Compute F and use p-value	Decision

Option A (SSE from ANOVA):

$$F = [(SSE_r - SSE_u) / J] \div [SSE_u / (n - k - 1)]$$

Option B (R^2 from Model Summary):

$$F = [(R^2_u - R^2_r) / J] \div [(1 - R^2_u) / (n - k - 1)]$$

Decision: Use the p-value (Sig.) from the SPSS ANOVA table.

Alternatively: if your computed $F > F$ -critical value $\rightarrow$ Reject H_0

SPSS: Running the Joint F-test (Restricted vs Unrestricted)

1. Run UNRESTRICTED model: Analyze $\rightarrow$ Regression $\rightarrow$ Linear...
2. Dependent: Y | Independent(s): X1, D_North, D_South, D_East (all variables)
3. Click OK. Record: SSE_u from ANOVA table (Residual row, Sum of Squares column)
4. OR record R^2_u from Model Summary table.
5. Run RESTRICTED model: Analyze $\rightarrow$ Regression $\rightarrow$ Linear...
6. Dependent: Y | Independent(s): X1 only (remove the variables being tested)
7. Click OK. Record: SSE_r (Residual SS) OR R^2_r from Model Summary.
8. Count J = number of variables you removed (e.g. 3 dummies $\rightarrow$ J = 3).
9. Compute F manually using the formula above. Compare p-value to 0.05.
10. TIP: n = Total df + 1 from the ANOVA table. k = number of IVs in unrestricted model.

Common Pitfalls

- J = number of RESTRICTIONS (variables REMOVED), not total variables.
- Restricted model has FEWER variables — its SSE is always LARGER.
- $n-k-1$ uses k from the UNRESTRICTED model.

1.7 Residual Plot Analysis

After running a regression, always inspect the residual plot (residuals on Y-axis, fitted values on X-axis). It reveals three types of violations:

What you see	Problem	Assumption violated
Random scatter centered on zero	None — model is fine	All assumptions hold ✓
Fan/cone shape — wider spread on one side	Heteroskedasticity	A2: $\text{Var}(\epsilon) = \sigma^2$ (constant)

Curved arc — residuals curve up then down	Non-linearity	A1: model correctly specified
One or two extreme isolated points	Outliers	Distort estimates — report them

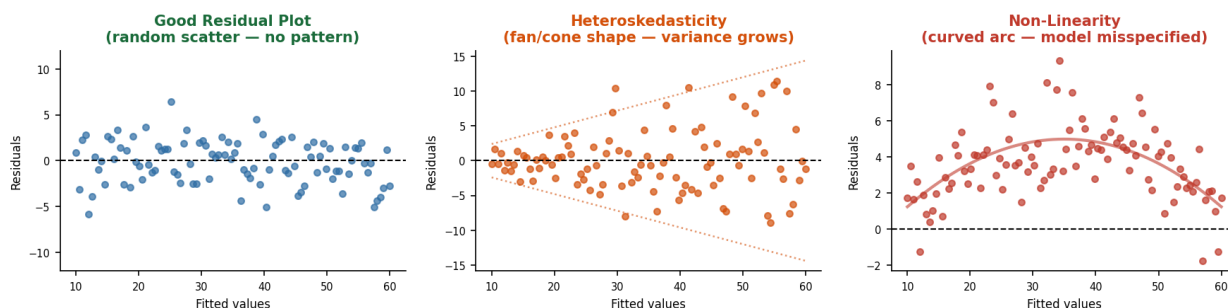


Figure 1.7 — Left: good residual plot — random cloud, no pattern. Centre: heteroskedasticity — fan shape, variance grows with fitted values. Right: non-linearity — curved arc, model is missing a term (e.g. X^2).

SPSS: Generating the Residual Plot

1. Analyze → Regression → Linear...
2. Dependent: Y | Independent(s): all predictors
3. Click Plots... button (bottom of dialog).
4. In the Plots dialog: set Y-axis to ZRESID (standardized residuals)
5. and X-axis to ZPRED (standardized fitted values). Click Continue.
6. Click OK. The scatter plot appears in the Output Viewer.
7. Inspect the plot using the three patterns in Figure 1.7 above:
8. • Random cloud → good | Fan shape → heteroskedasticity | Curved arc → non-linearity

Common Pitfalls

- Do not say 'residuals are not normal' from this plot alone. Normality requires a histogram or Q-Q plot.
- Heteroskedasticity → standard errors are wrong → t-tests and p-values are unreliable. Follow up with the White test.
- You do not need to quantify — describe what you SEE and name the assumption violated.

1.8 The White Test for Heteroskedasticity

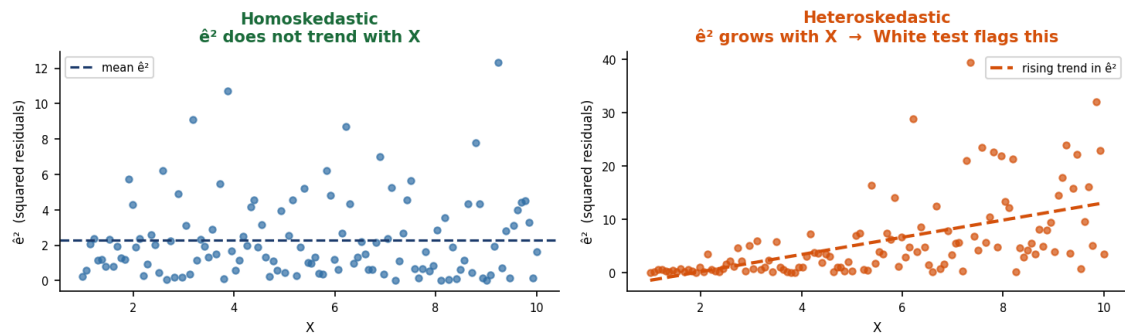


Figure 1.8 — Left: homoskedastic — squared residuals ($\hat{\varepsilon}^2$) show no trend with X . Right: heteroskedastic — $\hat{\varepsilon}^2$ rises with X . The White test auxiliary regression detects this rising pattern and flags it formally.

H_0 : homoskedasticity — $\text{Var}(\varepsilon) = \sigma^2$ (constant)

H_1 : heteroskedasticity — variance is NOT constant

Step 1: Run original model. Save residuals $\hat{\varepsilon}$ (Save → Unstandardized Residuals).

Step 2: Compute $\hat{\varepsilon}^2$ via Transform → Compute: $\text{RES_SQ} = \text{RES_1} ** 2$

Step 3: Auxiliary regression:

$\hat{\varepsilon}^2 = \alpha_0 + (\text{all original } X\text{'s}) + (X^2 \text{ for each continuous } X)$

+ $(X_i \times X_j \text{ for each pair of continuous } X\text{'s}) + u$

Do NOT square dummies. Do NOT include dummy×dummy cross-products.

Step 4: $W = n \times R^2_{\text{aux}} \sim \chi^2(\text{df})$

df = number of regressors in auxiliary regression (excluding constant)

Step 5: If $p < 0.05 \rightarrow \text{Reject } H_0 \rightarrow \text{heteroskedasticity present}$

SPSS: Running the White Test

1. Run original regression: Analyze → Regression → Linear... (Y on all X predictors).
2. Save residuals: click Save... → check Unstandardized (under Residuals) → Continue → OK.
3. SPSS creates variable RES_1 in the dataset.
4. Compute squared residuals: Transform → Compute Variable...
5. Target Variable: RES_SQ | Expression: RES_1 ** 2 | Click OK.
6. Run auxiliary regression: Analyze → Regression → Linear...
7. Dependent: RES_SQ
8. Independent(s): all original X's + X^2 for each continuous X + $X_i \times X_j$ cross-products
9. (Do NOT square dummies. Do NOT include dummy×dummy products.)

10. Click OK. Record R^2_{aux} from Model Summary and $n_{\text{aux}} = \text{Total df} + 1$ from ANOVA.
11. Compute: $W = n_{\text{aux}} \times R^2_{\text{aux}}$. $\text{df} = \text{number of IVs in auxiliary regression}$.
12. Decision: if $\text{Sig.} < 0.05$ on the overall F of the auxiliary regression $\rightarrow$ reject H_0 (heteroskedasticity).
13. OR: if $W > \chi^2_{\text{critical}}(\text{df}, 5\%) \rightarrow$ reject H_0 .

⚠ Common Pitfalls

- df is NOT 1. Example: 2 continuous X's $\rightarrow X_1, X_2, X_1^2, X_2^2, X_1 \times X_2 \rightarrow \text{df} = 5$.
- $D^2 = D$ for a 0/1 variable — never square dummies.
- Always state df explicitly in your answer: $W \sim \chi^2(5)$.

1.9 Omitted Variable Bias

$$\text{sign}(\text{bias on } \beta_X) = \text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$$

β_Z = true effect of the OMITTED variable Z on Y

$r(X, Z)$ = correlation between INCLUDED X and OMITTED Z

Bias $> 0 \rightarrow$ estimate is TOO HIGH (upward bias – overestimated)

Bias $< 0 \rightarrow$ estimate is TOO LOW (downward bias – underestimated)

$\text{sign}(\text{bias}) = \text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$			
$\beta_Z > 0$	$\beta_Z > 0$	$\beta_Z < 0$	$\beta_Z < 0$
UPWARD BIAS (estimate too high)	DOWNWARD BIAS (estimate too low)	DOWNWARD BIAS (estimate too low)	UPWARD BIAS (estimate too high)
$\beta_Z > 0$ $(+) \times (+) = +$	$\beta_Z > 0$ $(+) \times (-) = -$	$\beta_Z < 0$ $(-) \times (+) = -$	$\beta_Z < 0$ $(-) \times (-) = +$

Figure 1.9 — Sign grid for omitted variable bias. Top row: direction of β_Z (omitted variable's true effect on Y). Left column: sign of correlation between included X and omitted Z. Green = upward bias (estimate too high). Red = downward bias (estimate too low).

β_Z (effect of Z on Y)	$r(X, Z)$	Bias on β_X	Estimate is...
Positive (+)	Positive (+)	$+$ $\rightarrow$ upward bias	Too high
Positive (+)	Negative (–)	$-$ $\rightarrow$ downward bias	Too low
Negative (–)	Positive (+)	$-$ $\rightarrow$ downward bias	Too low
Negative (–)	Negative (–)	$+$ $\rightarrow$ upward bias	Too high

Worked Example

- Example: estimating effect of $\ln(\text{NOx})$ on housing values, omitting Indus (industry zone).
- $\beta_{\text{Indus}} < 0$ (more industry $\rightarrow$ lower house values).
- $r(\ln(\text{NOx}), \text{Indus}) > 0$ (high- NOx areas also tend to have more industry).
- Bias = $(-) \times (+) = \text{negative} \rightarrow$ estimate of $\beta_{\ln\text{NOx}}$ is TOO LOW (overestimates the negative effect).

1.10 VIF and Multicollinearity

$$\text{VIF}_j = 1 / (1 - R^2_j) \quad \text{Tolerance}_j = 1 / \text{VIF}_j$$

$R^2_j = R^2$ from regressing X_j on all other X 's (auxiliary regression)

Threshold: $\text{VIF} > 5 \rightarrow$ concerning $\text{VIF} > 10 \rightarrow$ severe

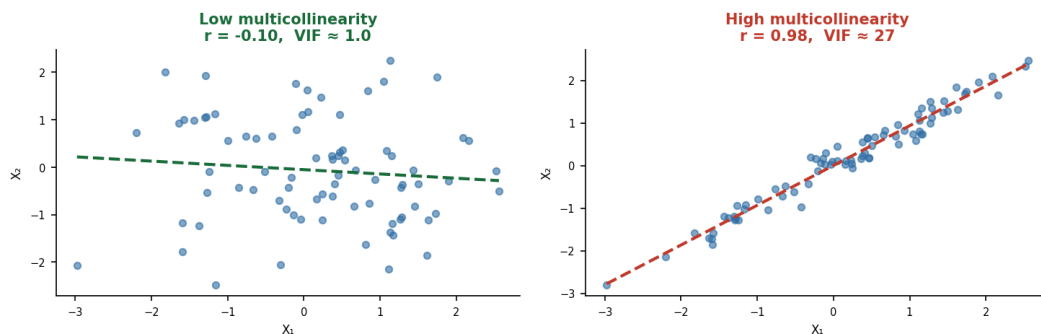


Figure 1.10 — Left: low multicollinearity — X_1 and X_2 are uncorrelated, $\text{VIF} \approx 1$. Right: high multicollinearity — X_1 and X_2 are nearly identical ($r \approx 0.95$), VIF is very large. When predictors are this correlated, SPSS cannot reliably separate their individual effects — coefficients and standard errors become unstable.

VIF	Tolerance	Interpretation
1.0	1.00	No multicollinearity — X uncorrelated with all others
1 to 5	0.20 to 1.00	Acceptable — moderate correlation, no real problem
5 to 10	0.10 to 0.20	Concerning — inspect estimates carefully
> 10	< 0.10	Severe — estimates are unreliable

SPSS: Getting VIF and Tolerance

1. Analyze $\rightarrow$ Regression $\rightarrow$ Linear...
2. Dependent: Y | Independent(s): all predictor variables
3. Click Statistics... button.
4. Check Collinearity diagnostics. Click Continue.
5. Click OK. In the Coefficients table, scroll right to find:

6. Tolerance column: should be close to 1.0 (low multicollinearity)
7. VIF column: flag any value above 5 as concerning, above 10 as severe.

1.11 Quick Reference — Phrasing → Method

If the question says...	It means...	Method
"Interpret the coefficient of $\ln(X)$ "	Effect of 1% change in X	$\beta/100$ units change in Y
"Test if Region is significant"	All dummies jointly	Joint F-test
"Is β_2 significantly different from zero?"	Individual significance	p-value from Coefficients table
"Test for heteroskedasticity"	Unequal error variance	White test: $W = n \times R^2_{\text{aux}}$
"What is the bias if Z is omitted?"	Direction of bias	$\text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$
"VIF = 2.5 — what does this mean?"	Multicollinearity severity	Compare to threshold of 5
"Write the theoretical model"	Full symbolic equation	$Y = \beta_0 + \beta_1 X_1 + \beta_2 D_1 + \dots$

PART 2 — TIME SERIES REGRESSION

2.1 The Three-Step Framework — Always in This Order

The Three-Step Framework — never skip or reorder

- STEP 1 — STATIONARITY: Dickey-Fuller test on every variable. Non-stationary series must be differenced before use.
- STEP 2 — FIT THE MODEL: Static, DL(q), or AR(1) depending on the question. For DL(q) use top-down lag selection.
- STEP 3 — CHECK AUTOCORRELATION: LMSC test on model residuals.

2.2 Step 1 — Stationarity and the Dickey-Fuller Test

Before running any time series regression you must confirm that every variable is stationary. The three plots below show what stationary and non-stationary series look like in a time plot — this is how you choose the DF version.

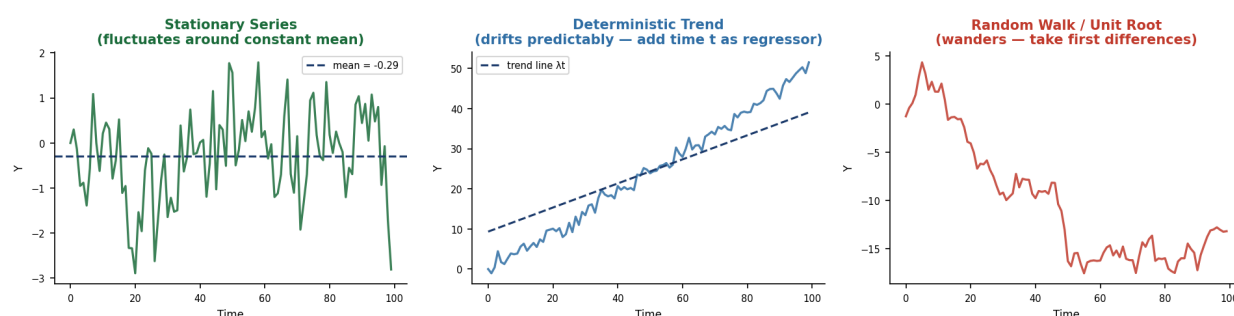


Figure 2.2a — Left (green): stationary series — fluctuates around a constant mean. Centre (blue): deterministic trend — drifts upward predictably, returns to trend after shocks (fix: add t as regressor). Right (red): random walk / unit root — wanders unpredictably, variance grows (fix: take first differences ΔY).

DF Version	Regression run	When to use
Version 1 — no trend	$\Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t$	Time plot shows NO visible upward or downward trend
Version 2 — with trend	$\Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$	Clear trend in time plot OR when unsure (safer default)

$H_0: \alpha_1 = 0$ (unit root — series is NON-STATIONARY)

$H_1: \alpha_1 < 0$ (series IS stationary) ← one-sided test

Special critical values at 5% — NEVER use ± 1.96 here:

Version 1 (no trend): -2.86

Version 2 (with trend): -3.41

Decision: t on Y_{t-1} MORE negative than CV $\rightarrow$ Reject $H_0 \rightarrow$ STATIONARY

t on Y_{t-1} LESS negative (closer to 0) $\rightarrow$ Fail to reject $\rightarrow$ UNIT ROOT

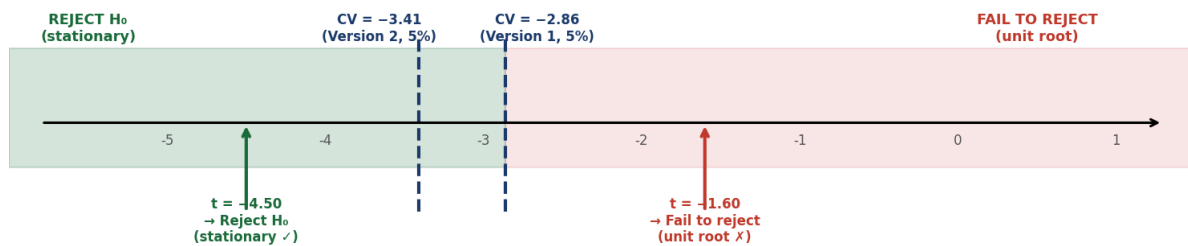


Figure 2.2b — DF decision number line. The shaded green region is the rejection zone (t more negative than the critical value). $t = -4.50$ falls in the rejection zone $\rightarrow$ stationary. $t = -1.60$ does not $\rightarrow$ unit root, must take first differences.

DF result	Action	Next step
Stationary (reject H_0)	Use series in levels	Proceed to model
Non-stationary (fail to reject)	Take $\Delta Y = Y_t - Y_{t-1}$	Run DF again on ΔY
ΔY is stationary (reject H_0)	Series is $I(1)$ — use first differences	Proceed to model
ΔY still non-stationary	Rare — take second differences	Series may be $I(2)$

SPSS: Dickey-Fuller Test (Step 1 — Stationarity)

1. CREATE FIRST DIFFERENCE: Transform $\rightarrow$ Compute Variable...
2. Target Variable: DIFF_Y | Expression: $Y - \text{LAG}(Y, 1)$ | Click OK.
3. CREATE LAGGED LEVEL: Transform $\rightarrow$ Compute Variable...
4. Target Variable: LAG_Y | Expression: $\text{LAG}(Y, 1)$ | Click OK.
5. RUN DF REGRESSION (Version 1 — no trend):
6. Analyze $\rightarrow$ Regression $\rightarrow$ Linear...
7. Dependent: DIFF_Y | Independent(s): LAG_Y
8. Click OK.
9. RUN DF REGRESSION (Version 2 — with trend):
10. Same as above but ALSO add your time variable (e.g. TIME or t) to Independent(s).
11. READ OUTPUT: Coefficients table $\rightarrow$ find row for LAG_Y $\rightarrow$ read the t column.
12. IGNORE the Sig. column! Compare t directly to -2.86 (V1) or -3.41 (V2).

13. t more negative than CV $\rightarrow$ Reject $H_0 \rightarrow$ STATIONARY.

14. t less negative (closer to 0) $\rightarrow$ Fail to reject $\rightarrow$ UNIT ROOT $\rightarrow$ difference the series.

15. IF UNIT ROOT: repeat steps 1–12 using DIFF_Y instead of Y (test the first difference).

⚠ Common Pitfalls

- NEVER use the standard t-table (± 1.96) for DF. Always use -2.86 (V1) or -3.41 (V2).
- The Sig. (p-value) SPSS shows for the LAG variable uses the wrong distribution — ignore it.
- Rejecting H_0 in DF means STATIONARY — opposite of most other tests.
- DF is one-sided: only reject when t is sufficiently NEGATIVE.

2.3 Step 2 — Which Model to Fit?

Model	Equation	When to use
Static	$Y_t = \alpha + \beta X_t + \varepsilon_t$	Question specifies it, or DL top-down gives $q = 0$
DL(q) — Distributed Lag	$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \dots + \beta_q X_{t-q} + \varepsilon_t$	Question asks for DL; delayed effects expected
AR(1) — Autoregressive	$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$	Question specifies AR(1); or persistent autocorrelation in DL

2.4 DL(q) — Top-Down Lag Selection

Create lag variables in SPSS: Transform $\rightarrow$ Compute Variable

$LAG1_X = LAG(X, 1)$ $LAG2_X = LAG(X, 2)$ $LAG3_X = LAG(X, 3)$

Top-down rule:

1. Fit DL(q_{max}). Look at p-value of the HIGHEST lag.
2. If $p > 0.05$: not significant $\rightarrow$ drop it $\rightarrow$ fit DL($q-1$).
3. If $p \leq 0.05$: significant $\rightarrow$ STOP. Current q is optimal.
4. Repeat until highest remaining lag is significant.

NEVER drop an intermediate lag: if lag 2 is significant,

keep lag 1 even if lag 1 is not significant — structure must be contiguous.

Total multiplier = $\beta_0 + \beta_1 + \beta_2 + \dots + \beta_q$ (long-run effect of permanent +1 in X)

Immediate effect = β_0 only

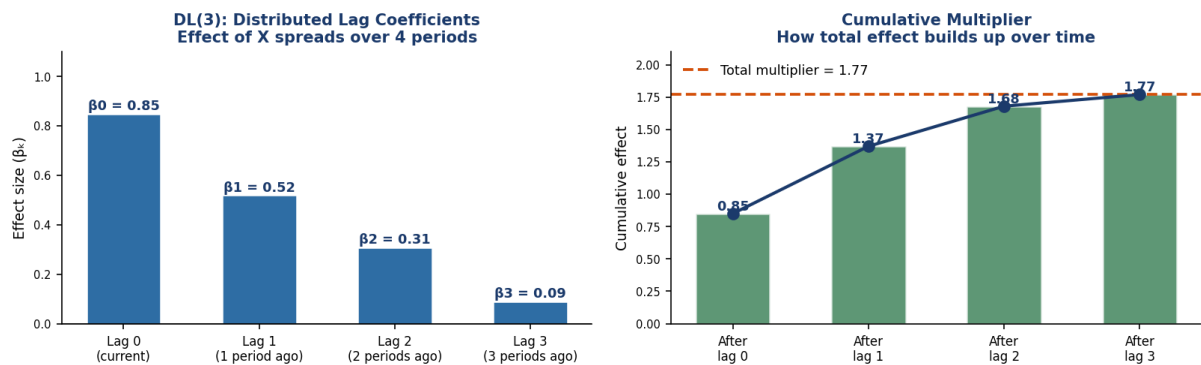


Figure 2.4 — Left: DL(3) coefficients. Each bar is the effect of X on Y at that lag. Right: cumulative multiplier — how the total effect builds up over time. The dashed line is the total multiplier (sum of all β coefficients). The lag-0 bar alone is the immediate effect; the total multiplier captures all delayed effects.

SPSS: DL(q) Top-Down Lag Selection (Step 2 — Model Fitting)

1. CREATE LAG VARIABLES (example for $q_{\max} = 3$):
2. Transform → Compute Variable... Target: LAG1_X Expression: LAG(X, 1) OK
3. Transform → Compute Variable... Target: LAG2_X Expression: LAG(X, 2) OK
4. Transform → Compute Variable... Target: LAG3_X Expression: LAG(X, 3) OK
5. FIT DL(3) — Analyze → Regression → Linear...
6. Dependent: Y | Independent(s): X, LAG1_X, LAG2_X, LAG3_X | Click OK.
7. Check Coefficients table: look at Sig. for LAG3_X (highest lag).
8. If Sig. > 0.05 → LAG3_X not significant → drop it → fit DL(2).
9. FIT DL(2) — repeat with Independent(s): X, LAG1_X, LAG2_X.
10. If Sig. of LAG2_X > 0.05 → drop → fit DL(1).
11. FIT DL(1) — repeat with: X, LAG1_X.
12. If Sig. of LAG1_X > 0.05 → drop → fit DL(0) (static model with X only).
13. STOP when the highest remaining lag is significant (Sig. ≤ 0.05).
14. TOTAL MULTIPLIER: sum all B values (X + LAG1_X + ... + LAGq_X) from the final model.

Common Pitfalls

- $q_{\max}$ is usually given in the question. Common default is 3 or 4.
- Each lag costs one observation. DL(3) with $n = 100$ gives 97 usable rows.
- Total multiplier $\neq \beta_0$. β_0 alone is the immediate effect only.

2.5 AR(1) — Autoregressive Model

Model: $Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$

β = immediate effect: 1-unit rise in X_t changes Y_t by β , holding Y_{t-1} fixed

γ = persistence: fraction of last period's Y that carries forward (requires $|\gamma| < 1$)

Total multiplier = $\beta / (1 - \gamma)$ (long-run effect of permanent +1 in X)

Example: $\beta = 1.0$, $\gamma = 0.65 \rightarrow$ Total multiplier = $1.0 / (1 - 0.65) = 2.86$

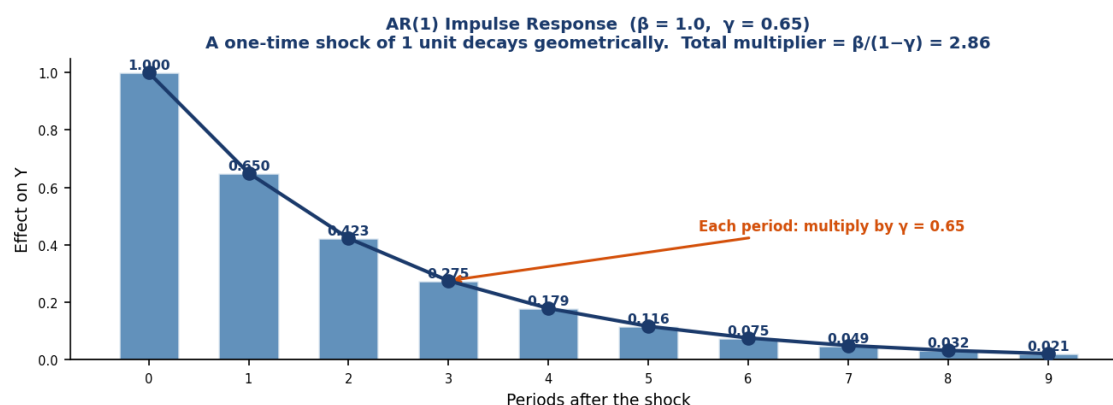


Figure 2.5 — AR(1) impulse response: what happens to Y in each period after a one-time shock of 1 unit in X_t . The immediate effect is $\beta = 1.0$. Each subsequent period, the effect is multiplied by $\gamma = 0.65$ — it decays geometrically. The sum of all bars equals the total multiplier $\beta/(1-\gamma) = 2.86$.

SPSS: Fitting the AR(1) Model

1. CREATE LAGGED Y: Transform → Compute Variable...
2. Target Variable: LAG_Y | Expression: LAG(Y, 1) | Click OK.
3. Analyze → Regression → Linear...
4. Dependent: Y | Independent(s): X and LAG_Y
5. Click OK.
6. Read Coefficients table:
7. B for X = β (immediate effect of a 1-unit rise in X this period)
8. B for LAG_Y = γ (persistence — must be between -1 and +1)
9. Compute total multiplier manually: $\beta / (1 - \gamma)$
10. Example: $\beta = 1.5$, $\gamma = 0.6 \rightarrow$ total multiplier = $1.5 / (1 - 0.6) = 3.75$
11. Then run the LMSC test (Section 2.6) on these residuals — autocorrelation biases AR(1) estimates.

△ Common Pitfalls

- AR(1) with remaining autocorrelation → coefficient estimates are BIASED. DL with autocorrelation → only INEFFICIENT (SE wrong, β correct). This is why always run LMSC after fitting AR(1).
- $|\gamma|$ must be < 1 for validity. If $|\gamma| \geq 1$ the series is non-stationary.
- "Persistence coefficient", "AR coefficient on lagged Y", "coefficient on Y_{t-1} " — all = γ .

2.6 Step 3 — LMSC Autocorrelation Test

Autocorrelation means the model's error in one period is correlated with the error in the previous period. This violates Assumption A3 and makes standard errors wrong. The plots below show what autocorrelated residuals look like compared to residuals with no autocorrelation.

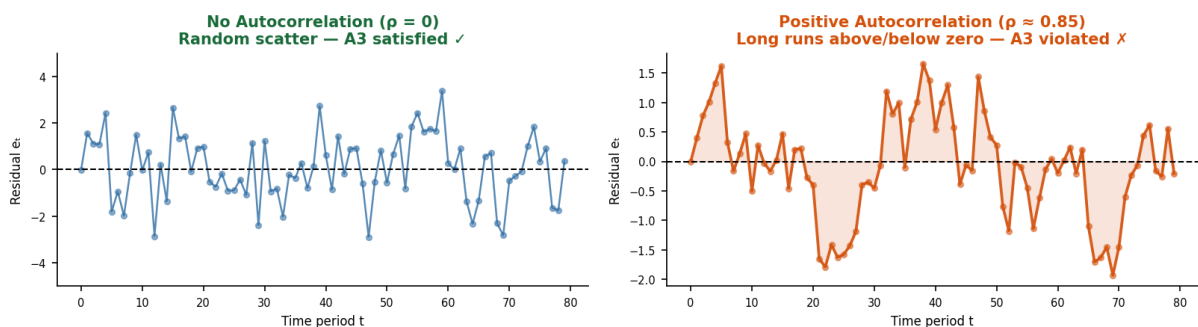


Figure 2.6a — Left: no autocorrelation — residuals jump randomly above and below zero. Right: positive autocorrelation — residuals form long smooth runs above zero, then long runs below zero. This is the pattern the LMSC test detects. If you see this in your residuals, A3 is violated.

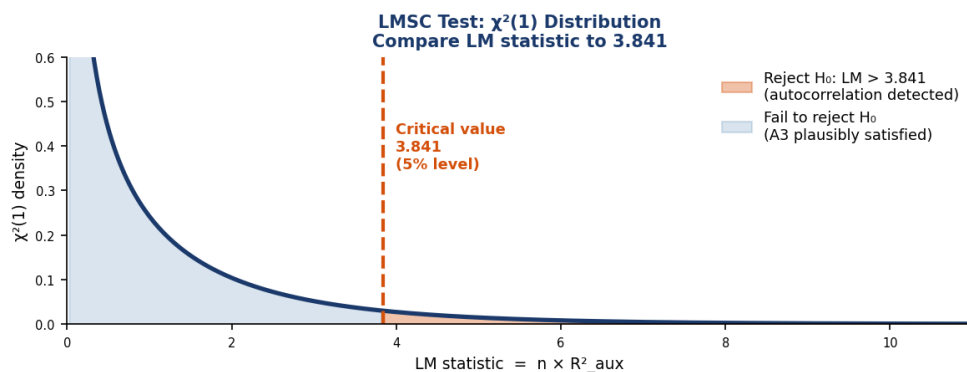


Figure 2.6b — $\chi^2(1)$ distribution for the LMSC test. The orange shaded region is the rejection zone ($LM > 3.841$). If your LM statistic falls in the orange zone, reject H_0 — autocorrelation is present.

H_0 : $\rho = 0$ (no autocorrelation — A3 satisfied)

H_1 : $\rho \neq 0$ (autocorrelation present — A3 violated)

Step 1: Fit original model. Save residuals: Save → Unstandardized Residuals → RES_1

Step 2: Create lagged residual: Transform → Compute → LAG_RES1 = LAG(RES_1, 1)

Step 3: Auxiliary regression: Dependent = RES_1

Independent = ALL original X's + LAG_RES1

Step 4: $n_{\text{aux}} = \text{Total df} + 1$ (from ANOVA table of auxiliary regression)

$LM = n_{\text{aux}} \times R^2_{\text{aux}} \sim \chi^2(1)$ under H_0

Step 5: If $p < 0.05$ (or $LM > 3.841$) → Reject H_0 → autocorrelation present

If $p \geq 0.05$ (or $LM \leq 3.841$) → Fail to reject → A3 plausibly satisfied

SPSS: LMSC Autocorrelation Test (Step 3 — Residual Check)

1. SAVE RESIDUALS from your original model:
2. Analyze → Regression → Linear... (run your fitted model)
3. Click Save... → check Unstandardized under Residuals → Continue → OK.
4. SPSS creates variable RES_1 in the dataset.
5. CREATE LAGGED RESIDUAL: Transform → Compute Variable...
6. Target Variable: LAG_RES1 | Expression: LAG(RES_1, 1) | Click OK.
7. RUN AUXILIARY REGRESSION: Analyze → Regression → Linear...
8. Dependent: RES_1
9. Independent(s): ALL original X variables + LAG_RES1
10. (e.g. if original model had X and LAG1_X: use X, LAG1_X, LAG_RES1 as IVs)
11. Click OK.
12. READ FROM OUTPUT:
13. Model Summary: record R^2_{aux}
14. ANOVA table: $n_{\text{aux}} = \text{Total df} + 1$ (the Total row, df column, add 1)
15. COMPUTE: $LM = n_{\text{aux}} \times R^2_{\text{aux}}$
16. DECISION: if $LM > 3.841$ OR overall model Sig. < 0.05 → Reject H_0 → autocorrelation present.
17. Otherwise: Fail to reject H_0 → A3 plausibly satisfied.

Common Pitfalls

- The auxiliary regression MUST include ALL original X predictors + lagged residual.
- Durbin-Watson (DW) is INVALID when Y_{t-1} appears as a regressor. Always use LMSC.
- "Test for autocorrelation", "check A3", "Breusch-Godfrey test", "LMSC" — all the same.

2.7 Critical Pitfall: Rejecting H_0 Means Opposite Things!

Test	H ₀ means...	Reject H ₀ means...	Fail to reject means...
Dickey-Fuller	Series HAS unit root (non-stationary)	Series IS stationary ✓	Non-stationary — difference it
LMSC	NO autocorrelation (A3 OK)	Autocorrelation present — fix model	A3 plausibly satisfied ✓
t-test	$\beta = 0$ (no effect)	Variable is significant	Cannot conclude variable matters
Joint F-test	All $\beta = 0$ (group no effect)	Group jointly significant	Group not significant
White test	Homoskedasticity (A2 OK)	Heteroskedasticity — A2 violated	A2 plausibly satisfied ✓

2.8 Deterministic vs. Stochastic Trends — Different Fix!

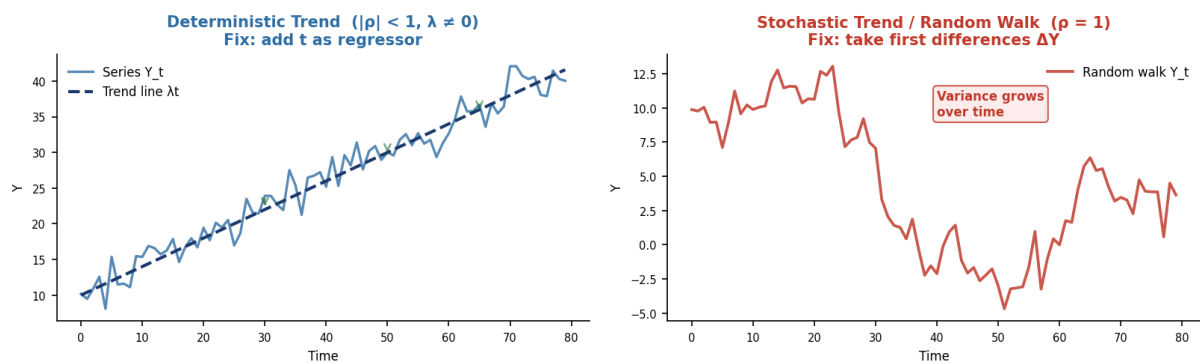


Figure 2.8 — Left (blue): deterministic trend — the series drifts upward predictably along the dashed trend line. After a shock it returns to the trend (green arrows). Fix: add t as a regressor. Right (red): stochastic trend / random walk — the series wanders erratically, never returning to a fixed level. Variance grows over time. Fix: take first differences ΔY . Applying the wrong fix makes things worse.

Feature	Deterministic trend	Stochastic / Random walk
Cause	λt term (mean grows linearly)	$\rho = 1$ (shocks are permanent)
Variance	Constant	Grows without bound
After a shock	Returns to trend	Never recovers
Correct fix	Add t as regressor	Take first differences ΔY
Wrong fix → consequence	First-differencing → artificial negative autocorrelation	Detrending does not remove a random walk

2.9 Time Series — Question Phrasings Reference

If the question says...	It means...	Method
"Is the series stationary?"	Does it have a unit root?	Dickey-Fuller test
"Test for a unit root"	Is $\rho = 1$?	Dickey-Fuller test

"Levels or differences?"	Stationarity property?	Run DF, then decide
"Determine optimal lag length"	Best q for DL(q)?	Top-down procedure
"Long-run effect of X on Y"	Total multiplier	DL: $\sum \beta_k$ AR: $\beta/(1-\gamma)$
"Immediate effect of X on Y"	Impact multiplier	β_0 only
"Test for autocorrelation"	Is A3 violated?	LMSC: LM = n × R ² _aux
"Is the model well-specified?"	Check all assumptions	LMSC + residual plot
"Series is I(1)"	Unit root in levels, stationary in differences	Use first differences
"Spurious regression risk"	Non-stationary series in levels	Run DF first
"Disturbance carries memory"	Autocorrelation present	LMSC test

2.10 Full SPSS Workflow

Step	SPSS action	Record
1a. Plot	Graphs → Chart Builder → Line → drag Y and TIME to axes	Does it trend? Wander?
1b. DF on levels	Compute DIFF_Y = Y – LAG(Y,1) and LAG_Y = LAG(Y,1). Regression: DV=DIFF_Y, IV=LAG_Y [+ TIME for V2]	t on LAG_Y vs –2.86/–3.41
1c. DF on differences	Compute DIFF2_Y = DIFF_Y – LAG(DIFF_Y,1) and LAG_DY = LAG(DIFF_Y,1). Regression: DV=DIFF2_Y, IV=LAG_DY [+ TIME]	t on LAG_DY vs –2.86/–3.41
2a. Lag variables	Transform → Compute: LAG1_X = LAG(X,1), LAG2_X = LAG(X,2)...	Obs lost = q
2b. Fit model	Regression with chosen variables	All β , R ² , n
3a. Save residuals	Regression → Save → Unstandardized Residuals	RES_1 created
3b. Lag residual	Transform → Compute: LAG_RES1 = LAG(RES_1, 1)	LAG_RES1 created
3c. Auxiliary reg.	Regression: DV=RES_1, IV=all original X's + LAG_RES1	R ² and n from output
3d. LMSC	LM = n_aux × R ² _aux. n_aux = Total df + 1 from ANOVA	Compare to 3.841

PART 3 — MASTER PITFALL & TRICK PHRASING LIST

3.1 The Top 15 Mistakes — and How to Avoid Them

1. Using p-value for DF test	SPSS's Sig. for the LAG variable uses the wrong distribution. Always compare the t-statistic directly to -2.86 (V1) or -3.41 (V2).
2. Reject H_0 in DF = non-stationary	OPPOSITE: rejecting H_0 in the DF test means the series IS stationary. H_0 is the unit root (non-stationarity).
3. Creating k dummies instead of k-1	For k categories, create k-1 dummies. Including all k causes the dummy trap (perfect multicollinearity). Always omit one reference category.
4. Using standard t-table for DF	The DF t-statistic does not follow a standard t-distribution. ± 1.96 does not apply. Use -2.86 or -3.41 only.
5. Missing X's in LMSC auxiliary regression	The auxiliary regression regresses residuals on ALL original X's PLUS the lagged residual. Missing any original X gives the wrong R^2 and wrong LM.
6. Total multiplier = β_0 only	DL(q): total multiplier = $\beta_0 + \beta_1 + \dots + \beta_q$ (ALL lag coefficients summed). AR(1): total multiplier = $\beta/(1-\gamma)$. β_0 alone is only the immediate effect.
7. Using DW for AR(1) models	Durbin-Watson is INVALID when Y_{t-1} appears as a regressor. Always use LMSC.
8. Differencing when trend is deterministic	Only difference when DF confirms a unit root. If trend is deterministic ($ \rho < 1$, $\lambda \neq 0$), add t as regressor instead. Wrong fix creates artificial autocorrelation.
9. Dropping intermediate insignificant lags	Top-down: only drop the HIGHEST lag while insignificant. Never drop lag 1 while keeping lag 2 — lag structure must be contiguous.
10. Squaring dummies in the White test	$D^2 = D$ for a 0/1 variable — squaring adds nothing. Only square continuous variables. Exclude dummy×dummy cross-products.
11. Confusing significant with large	A coefficient can be statistically significant ($p < 0.05$) but economically negligible. Always comment on both significance AND the size/direction of the effect.
12. Wrong sign for omitted variable bias	Bias = $\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$. Two negatives = positive bias. Use the sign grid if in doubt.

13. Ignoring reference category in interpretation	Coefficient on D_North means "compared to the reference group". Always state the reference category explicitly.
14. Using levels when both series are I(1)	If both Y and X have unit roots, always use first differences (unless cointegration is established — rare at introductory level).
15. Wrong n in LM formula	The auxiliary regression loses one observation for the lagged residual. Use $n_{\text{aux}} = \text{Total df} + 1$ from the ANOVA table (not the original n).

3.2 Trick Phrasings — Same Question, Different Words

What the question writes	What it actually means
"Is X relevant to explaining Y?"	"Is β_X significantly different from zero?" → p-value test
"Does adding Region improve the model?"	"Are the region dummies jointly significant?" → joint F-test
"Baseline group" / "comparison group"	"Reference category" — the omitted dummy group
"The disturbance carries memory"	"Autocorrelation is present" → LMSC test
"Series wanders without returning"	"Series has a unit root" → DF test, then difference
"Spurious correlation risk"	"Series is non-stationary" → run DF before modelling
"Long-run equilibrium effect"	"Total multiplier" → $\sum \beta_k$ or $\beta/(1-\gamma)$
"Impact effect of a price shock"	"Immediate effect" → β_0 only
"Error variance is not constant"	"Heteroskedasticity" → White test
"Predictors are too correlated"	"Multicollinearity" → VIF and Tolerance
"Coefficient on lagged Y is 0.7"	" $\gamma = 0.7$ in AR(1)" → total multiplier = $\beta/(1-0.7)$
"Series is integrated of order one"	"I(1) — unit root in levels, stationary in differences"
"Misspecification bias"	"Omitted variable bias" → $\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$
"First-order autocorrelation"	" $\varepsilon_t = \rho\varepsilon_{t-1} + u_t$ " → run LMSC test

QUICK DECISION GUIDE

MLR — Method Selector

Question type	Method	Key formula / output
Interpret coeff. of $\ln(X)$	Linear-log rule	$\beta/100$ units per 1% rise in X

Test one coefficient	Individual t-test	p-value (Sig.) in Coefficients table
Test a group of variables	Joint F-test	$F = [(SSE_r - SSE_u)/J] / [SSE_u/(n-k-1)]$
Check equal error variance	White test	$W = n \times R^2_{aux} \sim \chi^2(df)$
Direction of missing-variable distortion	Omitted variable bias	$\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$
Predictor correlation severity	VIF / Tolerance	$VIF = 1/(1-R^2_{aux})$, threshold=5
Write equation with categories	Dummy variables	k-1 dummies, state reference

Time Series — Method Selector

Question type	Method	Key output / decision
Is series stationary?	Dickey-Fuller test	$t < -2.86$ (V1) or -3.41 (V2) → stationary
Series drifts with time	DF test → unit root?	Unit root → difference. Deterministic → add t
Choose number of lags	Top-down DL	Drop highest lag while $p > 0.05$
Long-run effect of X on Y	Total multiplier	DL: $\sum \beta_k$ AR: $\beta/(1-\gamma)$
Immediate effect of X on Y	Impact multiplier	β_0 only
Check for autocorrelation	LMSC test	$LM = n \times R^2_{aux}$. Compare to 3.841
Model contains Y_{t-1}	Use LMSC, not DW	DW invalid for AR models

✓ Exam Tips

- Always state H_0 and H_1 before every test.
- Always write a full conclusion sentence: 'Since $p = X < 0.05$, we reject H_0 and conclude...'
- For DF: compare t-statistic (NOT p-value) to -2.86 or -3.41 .
- For LMSC and White test: use p-value directly (standard χ^2 distribution applies).
- For individual t-tests and F-tests: use p-value from SPSS output.
- When unsure about stationarity: always use DF Version 2 (safer default).
- Total multiplier = LONG-RUN effect of a PERMANENT change in X.